CS-009 Trimming, Sanding & Finishing

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Title	Trimming, Sanding & Finishing of Cured Composites
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Supersedes	Verbal Dremel training

Approvals

Role	Name	Date
Author	Simon Starbuck (drafted with Claude, SN6 Resources)	2026-07-12
Reviewer	simon (reviewer agent)	
Approver (Composites Lead)		

Revision history

Rev	Date	Author	Description of change
A	2026-07-12	Claude	Initial outline
B	2026-07-28	Claude	Language pass: fewer em dashes for rhythm variety, no content change.
C	2026-08-28	Claude	Out of outline, engineering pass (CS-000 Rev C conventions). §7's six numbered lines became subsections; §4 and §5 written out as definitions and an equipment table. Every number was already in Rev B or its references. Figure 1 draws the sanding budget (the sacrificial ply) beside the thin-laminate failure that killed the catch cans, which is the whole argument of §7.2 in one image.

1. Purpose

Cutting and sanding cured CF is where we make our dust (health problem), our edges (quality problem), and, in SN5, some of our damage: we punctured catch cans sanding thin laminate (2026-05-27), and Dremel work on molds ate heads and marred surfaces. These are the rules for cutting, sanding, and coating cured parts.

2. Scope

Cured CF/glass parts: trimming to scribe lines, edge finishing, surface prep for bonding, clear-coating. Waterjet/CNC cutting of flat panels included at outline level.

3. References

Ref	Document	Where
R1	DP420 TDS (structural bonding prep)	`03 Datasheets/3m-scotch-weld-epoxy-adhesive-d`
R2	Catch-can puncture incident	FEB-COMP-PP-001 §PP-05 / minors
R3	CS-002 (sacrificial ply concept), CS-010 (don't sand chasing conductivity)	this series

4. Definitions

- **Sacrificial ply**: the 88 spread-tow outer layer specified in the stack (CS-002 §7.1.3) to absorb sanding. It is the sanding budget; structural plies are not.
- **Trim line**: the part's cut boundary, transferred from the mold's scribe lines at demold.
- **Keying**: abrading a bond face so adhesive has something to grip. 120-grit for DP420 (R1).

5. Equipment & Materials

Item	Spec / product	Notes
Rotary tool + wheels	Dremel, reinforced cut-off wheels	Plain wheels shatter in CF; reinforced only
Dust extraction	Shop extractor at the cut	Runs for every cut and every sand of cured laminate (§6)
Sandpaper	120 → 400 → 800+	120 for trim and keying, 400 for edges, 800+ for pre-clear-coat
Waterjet	Jacobs, DXF through the WO	Flat panels (§7.5)
Clear coat	Acrylic Urethane A/B (Lime Line, in stock)	§7.4; encapsulates the part label (CS-001 §7.5)
Structural adhesive	3M DP420	R1: 2:1, 20 min work life, handling 2 h, full cure 24 h @72 °F

6. Safety

- **CF dust is the team's worst routine exposure:** respirator (P100/dust) + dust extraction ON for any cutting/sanding of cured laminate; long sleeves; gloves (cut CF edges splinter).
- **Supervision rule (SN5, stands until the lead changes it): no cutting cured carbon without a second trained person present.**
- Waterjet per Jacobs shop rules.

7. Procedure

7.1 Trim

1. Trim no earlier than the FEB hold for the resin system (CS-008 §5); the WO's demould step gates this.
2. Mark the cut from the scribe-transferred trim line. Tape the cut path: the tape controls chips and shows the line under dust.
3. Cut with the extractor running and a second trained person present (§6). Cut outside the line and sand to it; a Dremel does not steer well enough to cut on it.

7.2 Sand

1. **Read the WO stack before sanding anything.** The sanding budget is the sacrificial ply and only the sacrificial ply (Figure 1). Exposed structural weave past the plan: stop, record it, assess against the WO qualityChecks.
2. Zones under 3 plies: **hand-sand only, count strokes, check often.** Machine-sanding thin laminate is how the catch cans died (R2).

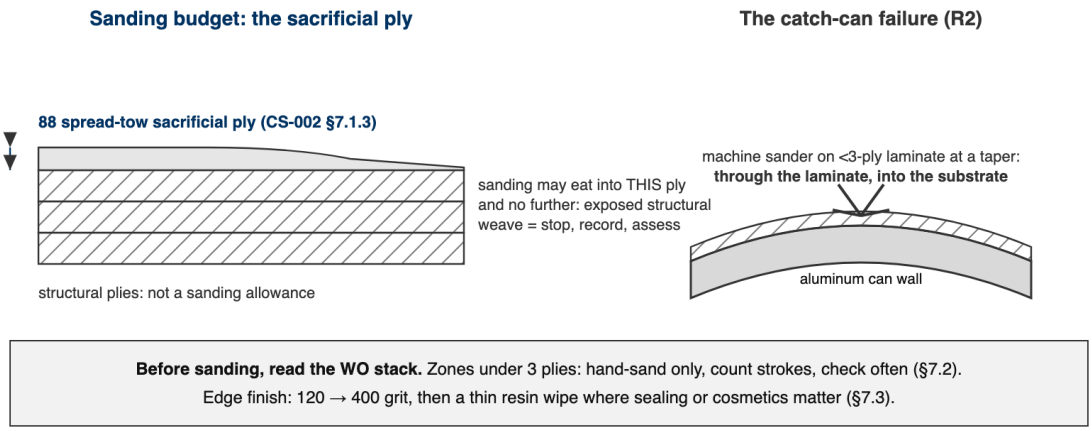


Figure 1: Left: the sacrificial ply is the sanding budget; structural plies are not an allowance. Right: the catch-can failure, a machine sander on thin laminate at a taper, through the skin and into the substrate.

3. Edges: 120 → 400 grit. Seal exposed edge fibers with a thin resin wipe where cosmetics or sealing matter; a raw CF edge wicks moisture and splinters into hands.

7.3 Bond prep (DP420)

1. Key both faces with 120-grit, then IPA wipe (never acetone near tooling board, CS-004 §7.1).
2. DP420 per R1: 2:1, 20 min work life, handling at 2 h, full cure 24 h at 72 °F. Clamp so squeeze-out forms a continuous fillet; that fillet is the §8 acceptance look.

[PHOTO: DP420 bond with continuous squeeze-out fillet: the §8 acceptance look]

7.4 Clear coat

1. Scuffed, IPA-wiped, label applied (CS-001 §7.5); then coat per product instructions and record coats on the WO.

7.5 Waterjet flat panels

1. The DXF comes from the requesting subteam **through the WO**, never a DM (the dashboard lesson, PP-04). Confirm the revision on the WO before cutting; Jacobs shop rules govern the machine.

8. Quality checks / acceptance criteria

Check	Target	Method	Record where
Trim to scribe	±1 mm of line	Visual/caliper	WO
No sand-through	Sacrificial ply intact / no exposed weave past plan	Visual	WO qualityChecks
Bond squeeze-out	Continuous fillet, no starved zones	Visual	WO + photo

9. Records

Trim/sand/bond/coat steps on the WO with operator + date; photos of finished edges and bonds.